# Sequences and Series

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Sequence limits**

Determine whether each sequence converges, and find the limit if it does:

**a** $a_n = \frac{3n+1}{n+2}$ Converges to $3$ (divide by $n$).

**b** $a_n = \frac{n^2}{2^n}$ Converges to $0$ (exponential growth dominates polynomial).

**c** $a_n = (-1)^n$ Diverges (oscillates between $-1$ and $1$).

(a) Converges to $3$ (divide by $n$).; (b) Converges to $0$ (exponential growth dominates polynomial).; (c) Diverges (oscillates between $-1$ and $1$).

**2** **[Foundational]** • **Target: Sequence limits**

Evaluate $\lim_{n\to\infty} \left(1+\frac{3}{n}\right)^n$.

This is the standard limit form giving $e^3$.

**3** **[Foundational]** • **Target: Series and partial sums**

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

Diverges — this is the harmonic series, a standard example of a divergent series with terms going to $0$.

**4** **[Foundational]** • **Target: Series and partial sums**

Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

$|r| = \frac{2}{5} < 1$: $S = \frac{3}{1-2/5} = 5$.

**5** **[Foundational]** • **Target: Series and partial sums**

Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. The partial sums telescope: $S_N = 1 - \frac{1}{N+1} \to 1$.

**6** **[Foundational]** • **Target: Monotonic and bounded sequences**

Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing $a_{n+1}$ and $a_n$.

$a_{n+1} - a_n = \frac{n+1}{n+2} - \frac{n}{n+1} = \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)} > 0$, so $a_{n+1} > a_n$: the sequence is increasing.

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**7** [Foundational] • Target: Monotonic and bounded sequences

Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by $1$) must converge, and find its limit.

By the Monotonic Convergence Theorem, a bounded, monotonic sequence converges. $\lim_{n\to\infty} \frac{n}{n+1} = 1$ (divide numerator and denominator by $n$).

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: More series practice

Evaluate the series $\sum_{n=1}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right]$ by splitting into two geometric series.

$\sum\left(\frac{1}{2}\right)^n = \frac{1/2}{1-1/2} = 1$. $\sum\left(\frac{1}{3}\right)^n = \frac{1/3}{1-1/3} = \frac{1}{2}$. Total $= 1 + \frac{1}{2} = \frac{3}{2}$.

**9** [Intermediate] • Target: More series practice

A ball dropped from 10 m rebounds to $60\%$ of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.

Total $= 10 + 2(10(0.6) + 10(0.6)^2 + \cdots) = 10 + 2 \cdot \frac{6}{1-0.6} = 10 + 2(15) = 40$ m.

**10** [Intermediate] • Target: Recursive sequences

A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute $a_2$, $a_3$, and $a_4$.

$a_2 = \frac{2+6}{2} = 4$. $a_3 = \frac{4+6}{2} = 5$. $a_4 = \frac{5+6}{2} = 5.5$.

**11** [Intermediate] • Target: Recursive sequences

Assuming the recursive sequence above converges to a limit $L$, find $L$ by setting $L = \frac{L+6}{2}$ and solving.

$2L = L + 6$, so $L = 6$.

**12** [Intermediate] • Target: Repeating decimals as series

Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.

$S = \frac{45/100}{1-1/100} = \frac{45/100}{99/100} = \frac{45}{99} = \frac{5}{11}$.

**13** [Intermediate] • Target: Visualizing partial sums of a series

The graph shows the first six partial sums $S_n = \sum_{k=1}^{n} \left(\frac{1}{2}\right)^k$ of a geometric series. Use the pattern to conjecture the value the sums are approaching, and confirm it using the geometric series formula.

The partial sums appear to approach $1$. Using the geometric series formula: $S = \frac{1/2}{1-1/2} = 1$, confirming the visual pattern.

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**14** [Intermediate] • Target: Series with factorial and combined terms

Evaluate $\sum_{n=0}^{\infty} \frac{1}{2^n} - \sum_{n=0}^{\infty} \frac{1}{3^n}$ by evaluating each geometric series separately.

$\sum \frac{1}{2^n} = \frac{1}{1-1/2} = 2$. $\sum \frac{1}{3^n} = \frac{1}{1-1/3} = \frac{3}{2}$. Difference $= 2 - \frac{3}{2} = \frac{1}{2}$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 6.02 Optimization and Curve Sketching

A sequence is defined by $a_n = \frac{2n + (-1)^n}{n}$. Find $\lim_{n\to\infty} a_n$, explaining why the $(-1)^n$ term does not affect the limit.

$a_n = 2 + \frac{(-1)^n}{n}$. As $n \to \infty$, $\frac{(-1)^n}{n} \to 0$ (bounded numerator divided by an unbounded denominator), so $\lim_{n\to\infty} a_n = 2$.

**16** [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 7.01 Integration by Substitution

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{\sin n}{n}$, using the fact that $-1 \le \sin n \le 1$ for all $n$.

$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$ for $n \ge 1$. Since $\lim_{n\to\infty} \left(\pm\frac{1}{n}\right) = 0$, by the squeeze theorem $\lim_{n\to\infty} \frac{\sin n}{n} = 0$.

**17** [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 7.02 Integration by Parts

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{n!}{n^n}$, using the bound $0 < \frac{n!}{n^n} \le \frac{1}{n}$ for $n \ge 1$.

Since $0 \le \frac{n!}{n^n} \le \frac{1}{n}$ and $\lim_{n\to\infty} \frac{1}{n} = 0$, by the squeeze theorem $\lim_{n\to\infty} \frac{n!}{n^n} = 0$.

**18** [Challenging] • Target: Applications of infinite series • Cross-Lesson: 8.01 Area Between Curves

A patient takes a $100$ mg dose of medication every $8$ hours. Between doses, $75\%$ of the drug remaining is eliminated, so a fraction $r = 0.25$ carries over to just before the next dose. Using an infinite geometric series, find the long-run steady-state amount in the body immediately after a dose (assume doses accumulate indefinitely).

Immediately after the $n$th dose, the amount is $100 + 100r + 100r^2 + \cdots = \frac{100}{1 - 0.25} = \frac{100}{0.75} \approx 133.33$ mg in the long run.

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**19** [Challenging] • Target: Recursive sequences • Cross-Lesson: 8.02 Volumes of Revolution

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is $r$. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given $x$ and $a, b$ are the limits of integration. Find the value of $a$ and $b$, and the expression for $V_{\text{hole}}$ in terms of $r$. The question asks for the setup of the integral representing the volume of the material removed.

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as $r$. The volume of the material removed by the cylindrical.

**20** [Challenging] • Target: Repeating decimals as series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{1000}$?

The error after summing up to $n = 6$ is at most the next term, $\frac{1}{5040} < \frac{1}{1000}$, while stopping one term earlier leaves a bound $\frac{1}{720} \geq \frac{1}{1000}$. So 7 terms are needed.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Limits of Sequences

Find the limit of the sequence $a_n = \frac{3n^2 - 5n}{2n^2 + 7}$ as $n \to \infty$.

**a.** 3/2 ✓

**b.** 0

**c.** $\infty$

**d.** 3/7

Divide numerator and denominator by $n^2$: $\lim_{n\to\infty} \frac{3 - 5/n}{2 + 7/n^2} = \frac{3}{2}$.

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**22** [Foundational] • Target: Limits of Sequences

Determine if the sequence $a_n = \frac{(-1)^n}{n}$ converges, and if so, find its limit.

**a.** **Converges to** $0$ ✓

**b.** Diverges (oscillates)

**c.** Converges to $1$

**d.** Converges to $-1$

Since $|a_n| = 1/n \to 0$ as $n \to \infty$, the sequence converges to 0 by the Absolute Value Theorem for sequences.

**23** [Foundational] • Target: Limits of Sequences

Find the limit of the sequence $a_n = \left(1 + \frac{2}{n}\right)^n$ as $n \to \infty$.

**a.** $e^2$ ✓

**b.** $e$

**c.** $1$

**d.** $\infty$

Using the standard exponential limit $\lim_{n\to\infty} (1 + x/n)^n = e^x$, with $x = 2$ we get $e^2$.

**24** [Foundational] • Target: Geometric Series

Find the sum of the geometric series $\sum_{n=1}^{\infty} 3\left(\frac{2}{5}\right)^{n-1}$.

**a.** $5$ ✓

**b.** $15/2$

**c.** $3$

**d.** $15$

First term $a = 3$, common ratio $r = 2/5$. Since $|r| < 1$, sum $S = \frac{a}{1-r} = \frac{3}{1-2/5} = \frac{3}{3/5} = 5$.

**25** [Foundational] • Target: Geometric Series

Determine the convergence of $\sum_{n=0}^{\infty} 4\left(-\frac{3}{2}\right)^n$.

**a.** **Diverges because** $|r| = 3/2 \geq 1$ ✓

**b.** Converges to $-8/5$

**c.** Converges to $8/5$

**d.** Converges to $0$

The common ratio is $r = -3/2$. Since $|r| = 1.5 \geq 1$, the geometric series diverges.

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**26** [Foundational] • Target: Geometric Series

Express the repeating decimal $0.\overline{27} = 0.272727\ldots$ as a simplified fraction using a geometric series.

**a.** 3/11 ✓ **b.** 27/100

**c.** 9/33 **d.** 27/99 is 3/11

$a = 27/100, r = 1/100. S = \frac{27/100}{1-1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}.$

**27** [Foundational] • Target: Telescoping Series

Find the sum of the telescoping series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

**a.** 1 ✓ **b.** 1/2

**c.** 2 **d.** Diverges

Partial fractions: $\frac{1}{n} - \frac{1}{n+1}$. Partial sum

$S_k = (1 - 1/2) + (1/2 - 1/3) + \ldots + (1/k - 1/(k+1)) = 1 - \frac{1}{k+1} \to 1$ as $k \to \infty$.

**28** [Foundational] • Target: Telescoping Series

Find the sum of $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$.

**a.** 1 ✓ **b.** 0

**c.** ∞ **d.** $\sqrt{2}$

$S_k = (1 - 1/\sqrt{2}) + (1/\sqrt{2} - 1/\sqrt{3}) + \ldots + (1/\sqrt{k} - 1/\sqrt{k+1}) = 1 - 1/\sqrt{k+1} \to 1.$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Telescoping Series

Evaluate $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$.

**a.** **Diverges to ∞** ✓ **b.** 0

**c.** 1 **d.** $\ln 2$

$S_k = \sum_{n=1}^{k} [\ln(n+1) - \ln(n)] = \ln(k+1) - \ln(1) = \ln(k+1) \to \infty$ as $k \to \infty$.

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**30** [Intermediate] • Target: Divergence Test (nth-Term Test)

What does the nth-Term Test for Divergence state?

**a.** **If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges ✓**

**b.** If $\lim_{n\to\infty} a_n = 0$, then $\sum a_n$ converges

**c.** If $\sum a_n$ diverges, then $\lim_{n\to\infty} a_n \neq 0$

**d.** If $\lim_{n\to\infty} a_n = 0$, the test proves divergence

The test states: if the terms do not approach 0, the series MUST diverge. If $\lim a_n = 0$, the test is inconclusive.

**31** [Intermediate] • Target: Divergence Test (nth-Term Test)

Apply the Divergence Test to $\sum_{n=1}^{\infty} \frac{n}{2n+1}$.

**a.** **Diverges because $\lim_{n\to\infty} a_n = 1/2 \neq 0$ ✓**

**b.** Converges to $1/2$

**c.** Inconclusive

**d.** Converges by p-series

$\lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} \neq 0$. Since the limit of the terms is non-zero, the series diverges.

**32** [Intermediate] • Target: Divergence Test (nth-Term Test)

Why is the Divergence Test inconclusive for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

**a.** **Because $\lim_{n\to\infty} \frac{1}{n} = 0$, which does not guarantee convergence ✓**

**b.** Because $1/n$ is negative

**c.** Because the terms oscillate

**d.** Because it is a geometric series

$\lim_{n\to\infty} 1/n = 0$. Having terms approach 0 is a necessary condition for convergence, but not a sufficient one.

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**33** [Intermediate] • Target: Telescoping Series

The sequence of partial sums $S_n = 2 - 1/2^n$: this means $\sum a_n =$

**a.** 1

**b.** 3/2

**c.** 2 ✓

**d.** diverges

Applying Sequences and Series principles, $\lim S_n = 2$. Therefore, the verified correct option is $2$.

**34** [Intermediate] • Target: Divergence Test (nth-Term Test)

Dirichlet test: $\sum a_n b_n$ converges if $a_n$ monotone $\to 0$ and partial sums of $\sum b_n$ are:

**a.** convergent

**b.** zero

**c.** increasing

**d.** **bounded** ✓

Dirichlet: bounded partial sums + monotone $a_n \to 0$.

**35** [Intermediate] • Target: Sequence limits

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, $a_5$?

**a.** **17** ✓

**b.** 20

**c.** 15

**d.** 23

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n - 1)d$, where $a_1$ is the first term and $d$ is the common difference. Given $a_1 = 5$ and $d = 3$, we want to find the 5th term, $a_5$. Using the formula: $a_5 = a_1 + (5 - 1)d$ $a_5 = 5 + (4)(3)$ $a_5 = 5 + 12$ $a_5 = 17$ Therefore, the 5th term of the sequence is 17.

**36** [Intermediate] • Target: Series and partial sums

A monotone bounded sequence:

**a.** diverges

**b.** **converges** ✓

**c.** oscillates

**d.** is arithmetic

Applying Sequences and Series principles, Monotone convergence theorem. Therefore, the verified correct option is converges.

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**37** [Intermediate] • Target: Monotonic and bounded sequences

An increasing sequence bounded above:

**a.** diverges

**b.** oscillates

**c.** **converges** ✓

**d.** is arithmetic

Applying Sequences and Series principles, Monotone convergence theorem. Therefore, the verified correct option is converges.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: More series practice • Cross-Lesson: 8.01 Area Between Curves

Sum of infinite geometric series ($|r| < 1$): $\sum_{k=0}^{\infty} ar^k =$

**a.** $\frac{a}{r}$

**b.** $\frac{a}{1-r}$ ✓

**c.** $a(1-r)$

**d.** $\frac{a}{1+r}$

Applying Sequences and Series principles, $S = a/(1-r)$, $|r| < 1$. Therefore, the verified correct option is $\frac{a}{1-r}$.

**39** [Challenging] • Target: Recursive sequences • Cross-Lesson: 8.02 Volumes of Revolution

Sum of $n$ terms of geometric $1 + 2 + 4 + \cdots$: $S_n =$

**a.** $2^n - 1$ ✓

**b.** $2^n$

**c.** $n \cdot 2^n$

**d.** $2^{n+1}$

Applying Sequences and Series principles, $S_n = (1 - 2^n)/(1 - 2) = 2^n - 1$. Therefore, the verified correct option is $2^n - 1$.

**40** [Challenging] • Target: Repeating decimals as series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Geometric series $3 + 1 + 1/3 + 1/9 + \cdots$: first term $a = 3$, $r =$

**a.** $1/3$ ✓

**b.** $1/2$

**c.** $3$

**d.** $1/9$

Applying Sequences and Series principles, Ratio: $1/3 \div 1 = 1/3$. Therefore, the verified correct option is $1/3$.

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**41** [Challenging] • Target: Visualizing partial sums of a series • Cross-Lesson: 9.01 Separable Differential Equations

Geometric sequence: $a_n = a_1 \cdot r^{n-1}$. If $a_1 = 2, r = 3$, then $a_4 =$

**a.** 6

**b.** 18

**c.** **54** ✓

**d.** 162

Applying Sequences and Series principles, $a_4 = 2 \cdot 27 = 54$. Therefore, the verified correct option is 54.

**42** [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 1.01 Precalculus Review for Calculus

Squeeze theorem for sequences: if $a_n \le b_n \le c_n$ and $a_n, c_n \to L$, then $b_n \to$

**a.** $L$ ✓

**b.** $L/2$

**c.** 0

**d.** $2L$

Applying Sequences and Series principles, Squeeze theorem. Therefore, the verified correct option is $L$.

**43** [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

A sequence is bounded above if there exists $M$ such that $a_n \le M$ for all $n$. This bound is:

**a.** always 0

**b.** the limit

**c.** **not necessarily unique** ✓

**d.** equal to the supremum

The bound M is not unique (any larger M also works).

**44** [Challenging] • Target: Applications of infinite series • Cross-Lesson: 12.01 Convergence Tests

Geometric sequence with $a_1 = 5$, ratio $r = -1/2$: does it converge?

**a.** No

**b.** Yes, to $-5$

**c.** Yes, to $5$

**d.** **Yes, to** $0$ ✓

$|r| < 1 \Rightarrow a_n = 5(-1/2)^{n-1} \to 0$.

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**45** [Challenging] • Target: Limits of Sequences • Cross-Lesson: 12.02 Power Series and Taylor Series

The $n$-th term test: if $\lim_{n\to\infty} a_n e 0$, the series $\sum a_n$:

**a.** converges

**b.** oscillates

**c.** may converge

**d.** **diverges ✓**

$n$-th term test: non-zero limit ■ diverges.

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